

Assignment 1

Harivdan Shrihari Narayanshastri - 125M1K010 – ADVT Lab

Title: Implement basic data visualization plots: Line plots, Area plots, Histograms, Bar charts, Pie charts, Box plots, Scatter plots

Aim

To implement and study basic data visualization plots – **Line plot, Area plot, Histogram, Bar chart, Pie chart, Box plot and Scatter plot** – using a real-world Superstore sales dataset and to interpret business insights from the visual patterns.

Dataset

- **Source:** Kaggle – *Superstore Sales Dataset* (superstore-dataset-final).
- **File name used:** superstore.csv.
- **Important attributes:**
 - Order Date – date of each order
 - Region – Central, East, West, South
 - Category – Furniture, Technology, Office Supplies
 - Sales – sales amount for each order
 - Profit – profit/loss for each order
 - Quantity, Discount and other descriptive fields

Software and Libraries

- Python 3
- pandas for data handling
- matplotlib for plotting graphs

Code

```
python
```

```
# Assignment 1: Basic Data Visualization Plots
```

```
# Dataset: Superstore sales dataset from Kaggle (superstore-dataset-final)
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# 1. Load dataset (update path if needed)
```

```
df = pd.read_csv("superstore.csv", encoding="latin1")
```

```
# Optional: view columns
```

```
print(df.columns)
```

```
# Convert Order Date to datetime
```

```
if 'Order Date' in df.columns:
```

```
    df['Order Date'] = pd.to_datetime(df['Order Date'])
```

```
# ----- LINE PLOT: Sales over time -----
```

```
if 'Order Date' in df.columns and 'Sales' in df.columns:
```

```
    sales_by_date = df.groupby('Order Date')['Sales'].sum().sort_index()
```

```
    plt.figure(figsize=(10, 5))
```

```
    plt.plot(sales_by_date.index, sales_by_date.values, marker='o', linewidth=1)
```

```
    plt.title("Line Plot – Daily Sales Over Time")
```

```
    plt.xlabel("Order Date")
```

```
    plt.ylabel("Sales")
```

```
    plt.tight_layout()
```

```
    plt.show()
```

```
# ----- AREA PLOT: Sales over time -----
```

```
if 'Order Date' in df.columns and 'Sales' in df.columns:
```

```
    plt.figure(figsize=(10, 5))
```

```
    plt.fill_between(sales_by_date.index, sales_by_date.values,  
                    alpha=0.4, color='skyblue')
```

```
    plt.plot(sales_by_date.index, sales_by_date.values,  
            color='blue', linewidth=1)
```

```
    plt.title("Area Plot – Daily Sales Over Time")
```

```
plt.xlabel("Order Date")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()
```

----- HISTOGRAM: Distribution of Sales -----

if 'Sales' in df.columns:

```
plt.figure(figsize=(8, 5))
plt.hist(df['Sales'], bins=30, edgecolor='black')
plt.title("Histogram - Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```

----- BAR CHART: Sales by Category -----

if 'Category' in df.columns and 'Sales' in df.columns:

```
sales_by_cat = df.groupby('Category')['Sales'].sum().sort_values(ascending=False)
```

```
plt.figure(figsize=(8, 5))
plt.bar(sales_by_cat.index, sales_by_cat.values, color='steelblue')
plt.title("Bar Chart - Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.tight_layout()
plt.show()
```

----- PIE CHART: Sales share by Region -----

col_for_pie = 'Region' if 'Region' in df.columns else 'Category'

```
sales_by_region = df.groupby(col_for_pie)['Sales'].sum()
```

```

plt.figure(figsize=(7, 7))

plt.pie(
    sales_by_region.values,
    labels=sales_by_region.index,
    autopct='%1.1f%%',
    startangle=90
)

plt.title(f"Pie Chart – Sales Share by {col_for_pie}")
plt.tight_layout()
plt.show()

# ----- BOX PLOT: Profit distribution -----
value_col = 'Profit' if 'Profit' in df.columns else 'Sales'

plt.figure(figsize=(6, 5))
plt.boxplot(df[value_col].dropna(), vert=True)
plt.title(f"Box Plot – Distribution of {value_col}")
plt.ylabel(value_col)
plt.tight_layout()
plt.show()

# ----- SCATTER PLOT: Sales vs Profit -----
if 'Sales' in df.columns and 'Profit' in df.columns:
    plt.figure(figsize=(8, 5))
    plt.scatter(df['Sales'], df['Profit'], alpha=0.5)
    plt.title("Scatter Plot – Sales vs Profit")
    plt.xlabel("Sales")
    plt.ylabel("Profit")
    plt.tight_layout()
    plt.show()

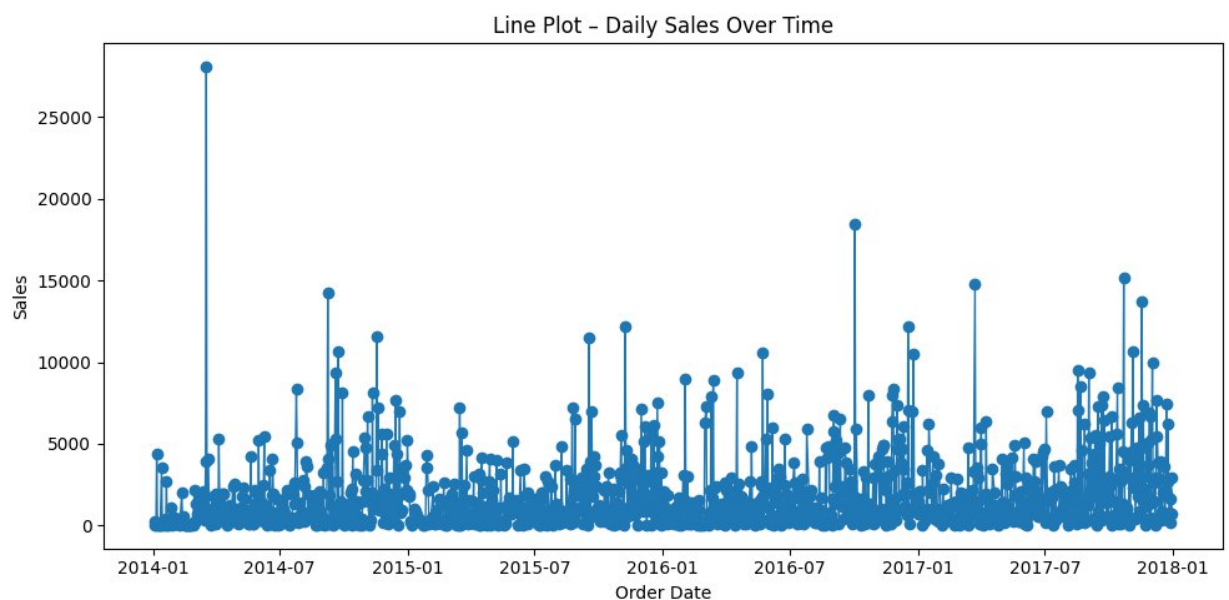
```

Procedure (write as steps)

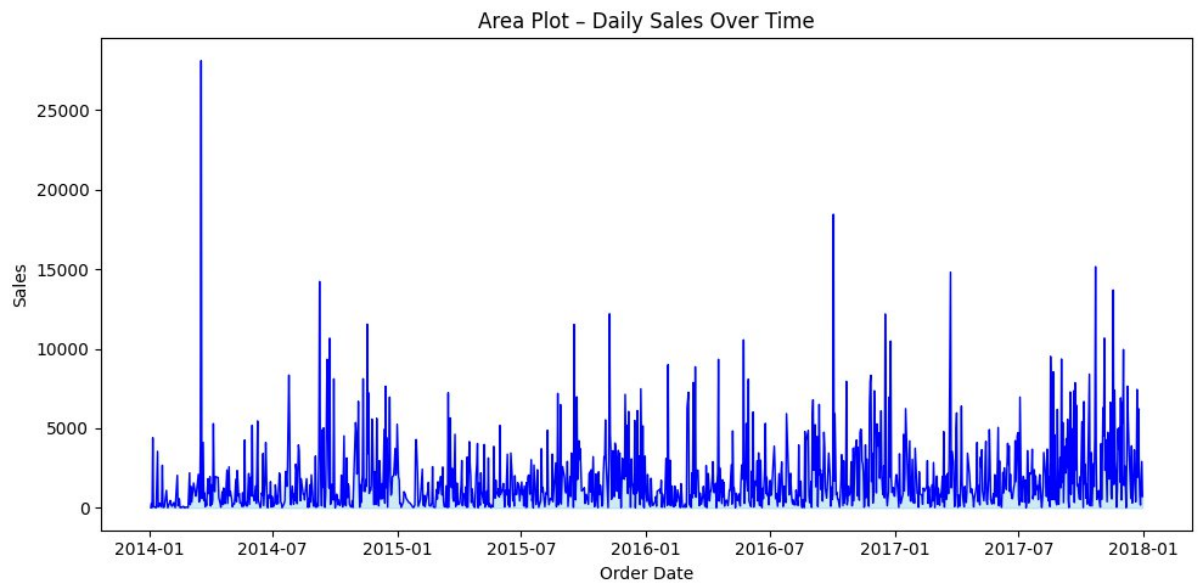
1. Downloaded the **Superstore Sales dataset** from Kaggle and saved it as superstore.csv in the working directory.
2. Imported the dataset into Python using `pandas.read_csv()` and verified the available columns.
3. Converted the Order Date field to datetime format for time-series analysis.
4. Grouped sales values by Order Date and plotted **Line** and **Area** charts to observe daily sales trends over the years.
5. Plotted a **Histogram** of Sales to visualize the distribution and frequency of different sales amounts.
6. Aggregated Sales by Category and created a **Bar chart** to compare performance of Furniture, Technology and Office Supplies.
7. Aggregated Sales by Region and plotted a **Pie chart** to show percentage contribution of Central, East, West and South regions to total sales.
8. Created a **Box plot** of Profit to study spread, median, quartiles and presence of outliers / loss-making orders.
9. Plotted a **Scatter plot** between Sales and Profit to check correlation between order value and profitability.

Output

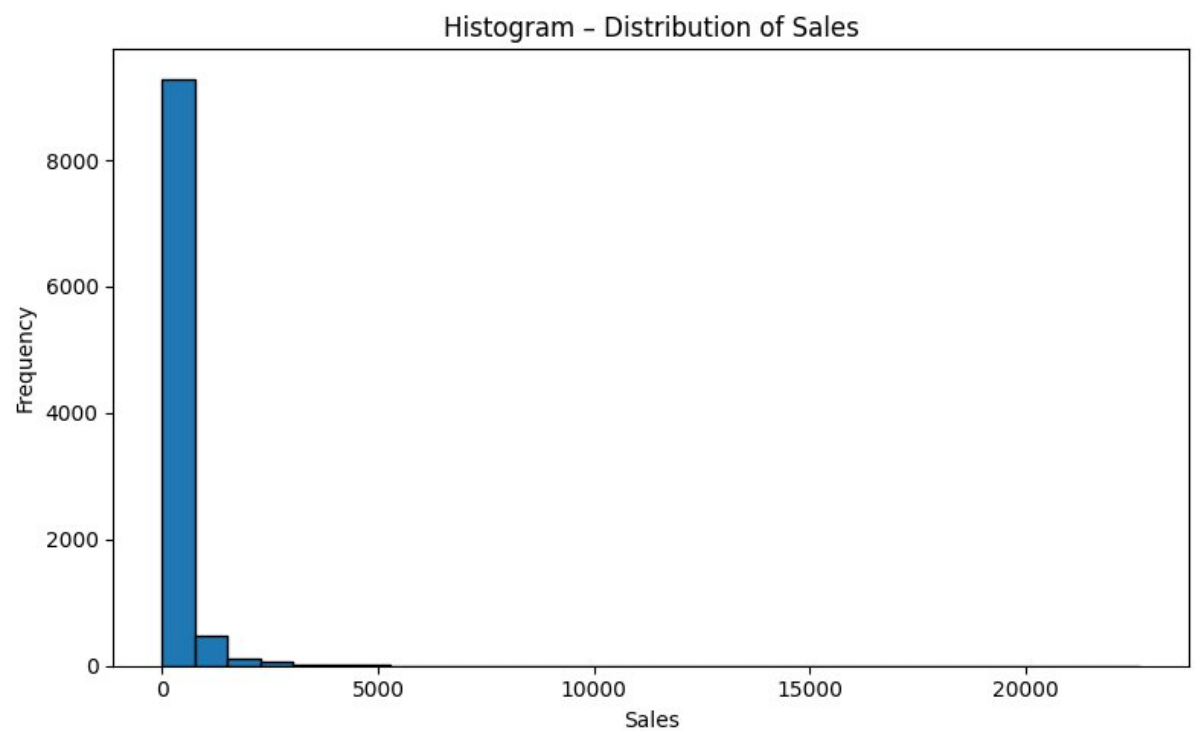
- **Line Plot – Daily Sales Over Time –**



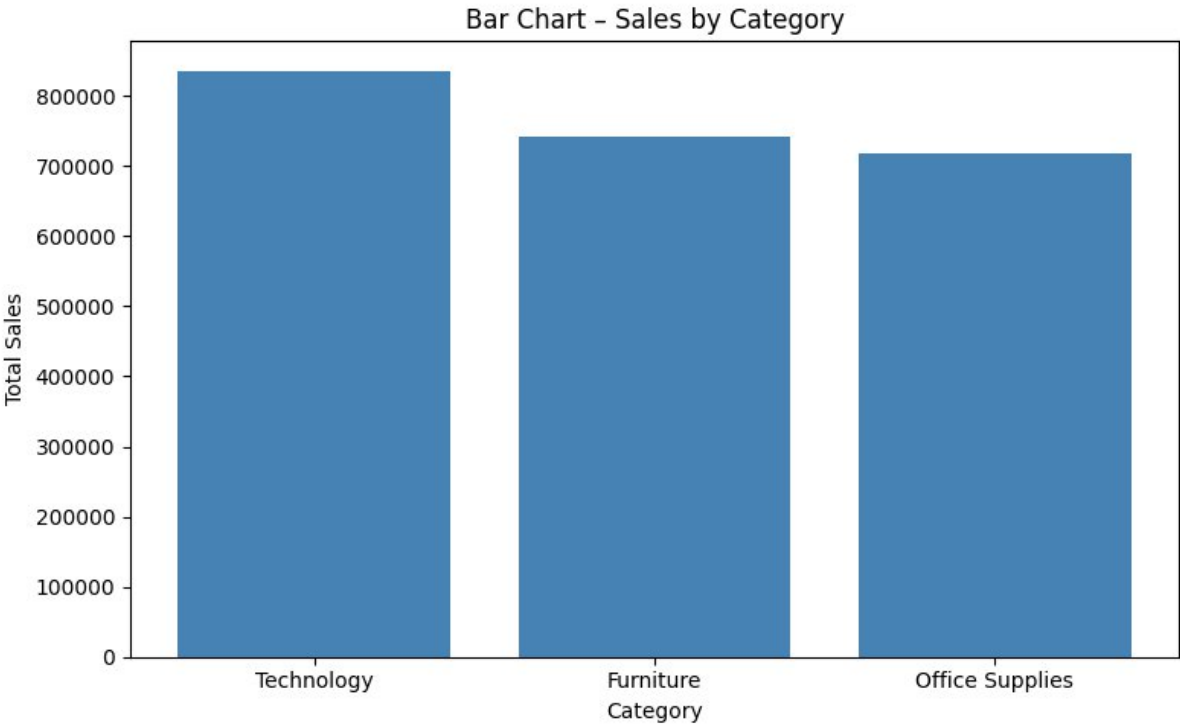
- **Area Plot - Daily Sales Over Time -**



- **Histogram - Distribution of Sales**

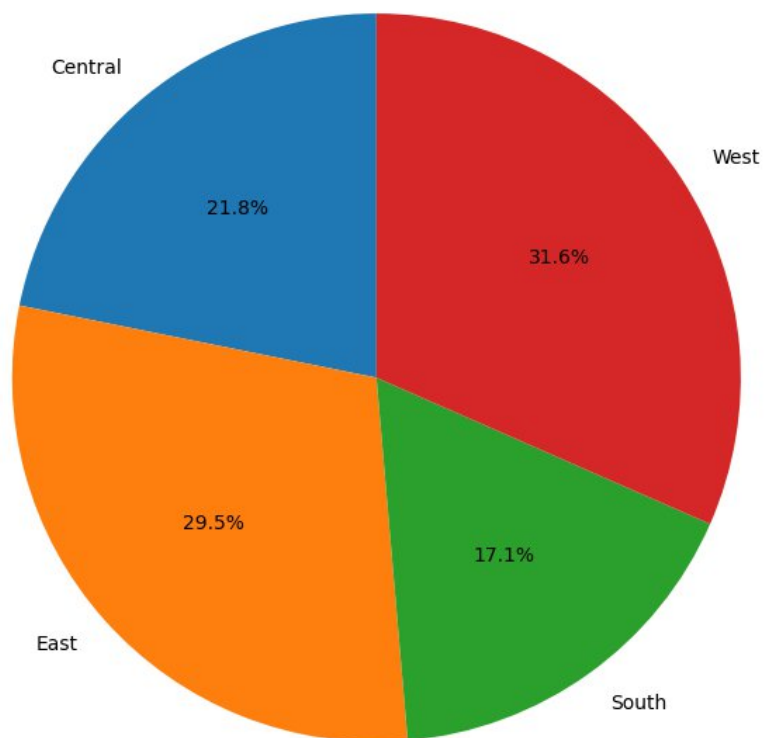


- Bar Chart - Sales by Category -



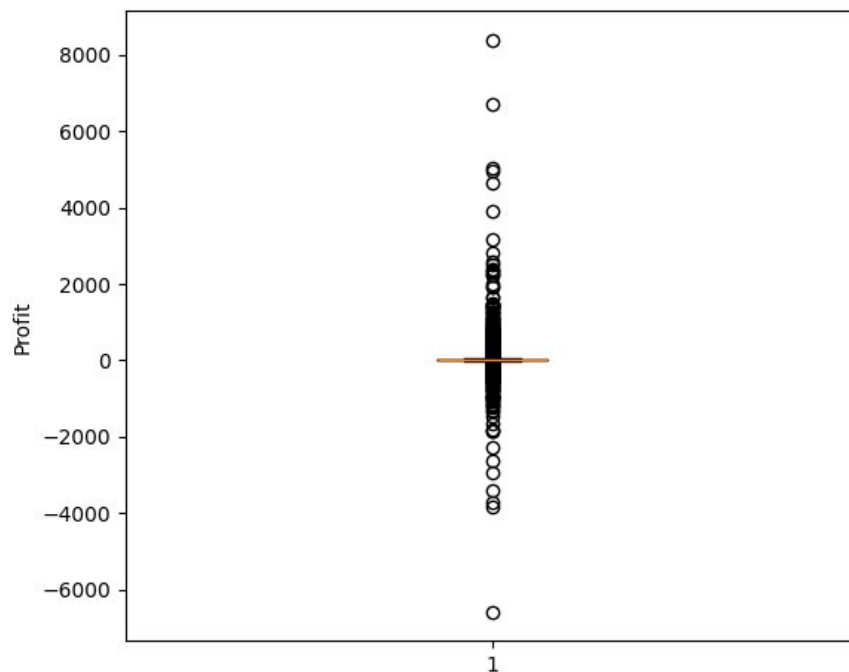
- **Pie Chart – Sales Share by Region –**

Pie Chart – Sales Share by Region

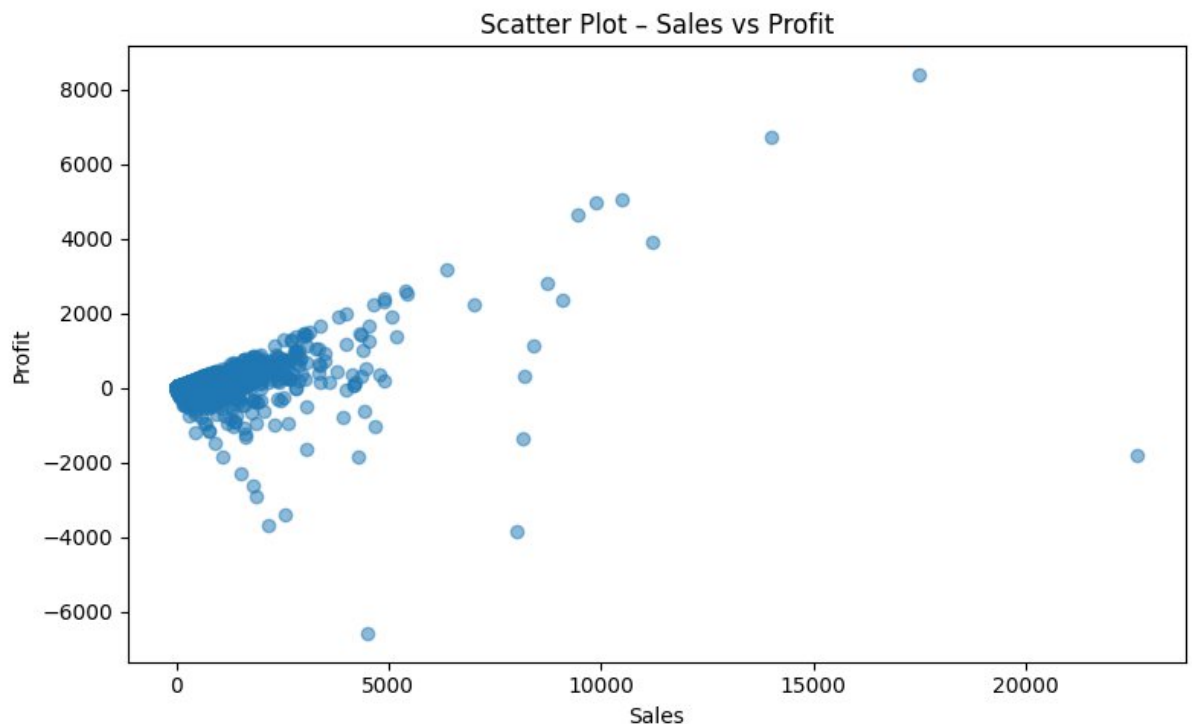


- **Box Plot – Distribution of Profit –**

Box Plot – Distribution of Profit



- **Scatter Plot – Sales vs Profit –**



Observations

Write these lines in your words:

1. The **line and area plots** show strong variation in daily sales over time with occasional very high spikes, indicating promotional periods or bulk orders in the Superstore data.
2. The **histogram** shows that most orders have relatively low sales amounts, while very high-value orders are rare, giving a right-skewed distribution.
3. The **bar chart** indicates that all three categories contribute significantly, with Technology having slightly higher total sales than Furniture and Office Supplies in this dataset.
4. The **pie chart** shows that the West and East regions contribute the largest share of total sales, while the South region contributes the least, suggesting regional differences in business volume.
5. The **box plot of Profit** reveals many outliers, including some negative values, indicating that certain orders result in heavy losses due to discounts or costs.
6. The **scatter plot of Sales vs Profit** shows an overall positive relationship between sales and profit, but also some high-sales orders with low or negative profit, highlighting the impact of discounting and pricing strategy.

Conclusion

Basic data visualization techniques such as line, area, histogram, bar, pie, box and scatter plots were successfully implemented on a real Kaggle Superstore dataset using Python and matplotlib.

These visualizations provide clear insight into sales trends, regional and category-wise performance, distribution of sales and profit, and the relationship between order value and profitability, which is essential for effective analysis and decision-making in advertising and data-driven marketing.